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Ideas to create a "Net Zero AI Data Center" AKA combating AI energy usage?

Posted by u/whereisthewifiplugtf • • 0 points • 6 comments

For context me and a couple of friends are participating in an AEC industry competition where essentially our goal is to create a new building/site which, in exact words, "significantly reduces its environmental impact / carbon footprint and respects the surrounding community" where we are able to conserve energy and excess heat created from technology and more specifically AI. That being said all of us have very basic and surface level of how AI works. (so PLEASE correct me if anything i say is wrong.)

There is no budget for this theoretical data center however because it is only one site/building it's not like we are adding/changing any laws or pushing popular opinion that will be able to significantly improve/support green energy usage or restrict AI usage

My ideas so far are

\- integrating types of renewable energy such as nuclear energy into the data center (how would this work? use the condensation stage of the water in the nuclear plant for cooling?)

\- copy China and create an underwater ocean data center (unoriginal so this is plan B for now)

\- rainwater collection and greywater usage for cooling so less freshwater is used

OR use a whole different liquid for cooling altogether

would really appreciate any ideas, thanks! This project is in its early early stages of development so I'll hopefully be able to introduce or integrate any suggestions I see

Comments

u/X7123M3-256 • 2 points • 2026-01-22 02:11 UTC

> integrating types of renewable energy such as nuclear energy into the data center (how would this work? use the condensation stage of the water in the nuclear plant for cooling?)

* Nuclear power plants are carbon neutral but are not renewable energy, they require uranium fuel which is a finite resource

* Nuclear power plants are very expensive, often take decades to build and are subject to very strict regulatory requirements. It really wouldn't make any sense for a data center to be powered by its own dedicated nuclear plant, it would make more sense to use renewable energy instead.

* The condensation stage of a power plant is where the hot steam coming from the turbine needs to be cooled back down so it will condense back into water. You cannot use the water that you use to cool the steam to then cool something else because after passing through the power plants cooling loop it is now hot. Nuclear power plants create a lot of heat and require a lot of cooling water - those giant concrete towers you probably think of when you think of a nuclear plant are cooling towers. You would need additional cooling water for the plant separate from that used to cool the data center.

> OR use a whole different liquid for cooling altogether

No sensible alternative, water is incredibly cheap, nontoxic and one of the most effective coolants you could use due to its high heat capacity. Anything else would be worse by pretty much any measure.

An idea that you are missing is to use the hot water for a useful purpose such as heating homes and businesses. This is already being done in some areas. In doing this, you are saving energy that would otherwise be used for heating by using the waste heat from the data center instead.

u/[deleted] • 1 points • 2026-01-21 05:28 UTC

[deleted]

u/Regular-Wrangler264 • 3 points • 2026-01-21 05:03 UTC

At the risk of being a Debby downer, I'm not sure you're grasping the amount of energy you're trying to offset.

A single rack of modern nVidia gpus can pull 150 kW. A datacenter has 15000+ racks. That's a potential 2,250,000 kW or 2.25 GW. A large datacenter can have 10x that number of racks, but will focus on the theoretical low end1600.

A whole nuclear powerplant only generates like 1.6 gW, so you need 2 of those.

Or about ~1400 wind turbines.

Or ~15 million solar panels.

For one datacenter.

Sources:
https://www.sunbirddcim.com/blog/how-much-power-does-nvidia-gb300-nvl72-need

https://www.racksolutions.com/news/blog/how-many-servers-does-a-data-center-have/

https://climate.mit.edu/ask-mit/how-many-wind-turbines-would-it-take-equal-energy-output-one-typical-nuclear-reactor

The condensation stage of a nuclear power plant is using fresh water to cool the steam. The water being condensed stays in the pipes to get turned into steam again. The water doing the cooling is evaporating out the top of the towers as steam.

The problem is the coolant needs to stay below boiling. Most electronics operate at a max of like 90 degrees Celsius, so the coolant needs to be significantly below that, as thermal transfer is proportionate to the temperature difference.

Therefore the ability to 'regenerate' electricity via high pressure steam and capture it is nearly impossible without some form of compressor / heat pump to concentrate it, which uses MORE electricity.

I'd focus on what you could use that heat for instead. Maybe something like municipal heating? Build apartments on top of datacenters or put the datacenters in city centers, pipe out the waste heat? But I'd bet that's only a fraction. As a whole house only uses like 30 kWH of electricity a day.

u/Concise_Pirate • 2 points • 2026-01-21 03:56 UTC

Of course you can always build a solar or nuclear or wind power plant just for the data center which makes it net zero. But there's really no logical reason to combine the two.

u/GESNodoon • 1 points • 2026-01-21 03:47 UTC

Create an ai for which the sole purpose is maintaining renewable energy, nuclear power plants and planting trees, I guess.

u/whereisthewifiplugtf • 1 points • 2026-01-21 03:47 UTC

I forgot to write this in the original post but an issue with the different water idea is that we would need a lot of different (large and winding) pipelines throughout the center to keep the waters seperate, along with the fact that the center needs flitered water for the inhabitants aka the workers working at the center. not very efficient space or money wise 😕

u/Nataliexxtyy • 2 points • 2026-01-21 03:28 UTC

Don’t worry about being surface level yet this is exactly how good projects start. The cooling and water ideas are especially promising.